

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 5: Instrument Landing System (ILS: LOC, GP & Markers)

Syllabus Source: CAE Oxford (pp. 169-216) | Pass Target: >= 90% AviationExam

1. THE ILS TRI-COMPONENT ARCHITECTURE

The Instrument Landing System provides precision lateral, vertical, and distance guidance during final approach:

ILS Subsystem	Frequency Band & Modulation Carrier	Beam Dimensions & Guaranteed Coverage
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Localizer (LOC)	VHF (108.10 to 111.95 MHz, odd tenths). 90 Hz (Left) / 150 Hz (Right) DDM.	Course width: ~5° (total 350 ft at threshold). Coverage: 35° each side to 17 NM; 10° each side to 25 NM.
Glide Path (GP)	UHF (329.15 to 335.00 MHz, paired with LOC). 90 Hz (Above) / 150 Hz (Below) DDM.	Slope: Standard 3.0°. Beam thickness: ~1.4° (0.7° above/below). Coverage: 8° each side to 10 NM. False slopes at 6° and 9°!
Marker Beacons	VHF 75.0 MHz (vertical fan beam).	Outer (OM: 400 Hz, blue dashes, 4-7 NM); Middle (MM: 1300 Hz, amber dot-dashes, 0.5-0.8 NM); Inner (IM: 3000 Hz, white dots).

2. GLIDE PATH RATE OF DESCENT (ROD) FORMULA

- Master ROD Formula (3.0° Glide Path): Rate of Descent (ft/min) = Groundspeed (knots) x 5 (e.g. at 140 kt GS: ROD = 140 x 5 = 700 ft/min!).
- Non-Standard Slope Formula: ROD (ft/min) = Groundspeed (knots) x Glide Angle (degrees) x 100 / 60.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Target VSI on 3-Degree ILS Glide

Problem: An airliner flies an ILS 3.0° glide path. Approach speed Vref is 140 kt, headwind is 20 kt (Groundspeed = 120 kt). What rate of descent on the VSI must be maintained? Step 1: Quick mental rule for a 3-degree path (5.2% gradient): - VSI (ft/min) = Groundspeed x 5 (or half groundspeed with a zero). Step 2: Calculate for GS = 120 kt: VSI = 120 x 5 = 600 ft/min. Step 3: If tailwind increases groundspeed to 150 kt: VSI = 150 x 5 = 750 ft/min required! Safety Rule: Rates of descent > 1,000 ft/min below 1,000 ft AGL violate stabilized approach criteria and require an immediate go-around.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Target Rate of Descent on ILS

SCENARIO (Everyday Airline Flying Drill): An airliner is established on an ILS approach with a 3.0° glide slope: - Target approach speed (V_{ref}) = 140 knots Indicated Airspeed - Headwind component on final approach = 20 knots QUESTION: What target Rate of Descent (ft/min) must the flight crew maintain on the vertical speed indicator (VSI)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the Groundspeed (GS): - Remember: Rate of descent depends on GROUND SPEED, not airspeed! - Groundspeed = TAS - Headwind - At sea level, TAS ~ IAS = 140 kt. - Groundspeed = 140 kt - 20 kt = 120 knots! Step 2: Recall the Master 3.0° Glide Path Formula: - Rate of Descent (ft/min) = Groundspeed (kt) x 5! Step 3: Calculate Target ROD: - ROD = 120 knots x 5 = 600 ft/min! FINAL ANSWER: Target Rate of Descent is 600 ft/min. If headwind decreases, groundspeed rises and ROD must increase proportionally.

[!] EXAM TRAP (AVIATIONEXAM) - False Glide Path Trap

- False glide paths are generated at HARMONIC MULTIPLES (typically $2 \times 3^\circ = 6^\circ$, and $3 \times 3^\circ = 9^\circ$)! - The first false glide path at 6° has NORMAL SENSING (90 Hz above, 150 Hz below)! - If intercepted from above, you will descend at double the normal rate (1,400 ft/min)!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: At what angle does the lowest FALSE GLIDE PATH occur on a standard 3.0° ILS installation?**

- [A]: 4.5°
- [B]: At DOUBLE the nominal glide path angle: 6.0 DEGREES
- [C]: 9.0°
- [D]: 1.5°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. False glide paths occur at multiples of the true angle; the lowest false path is at twice the true slope ($2 \times 3^\circ = 6^\circ$).

Q2: In which frequency band does the ILS GLIDE PATH transmitter operate?

- [A]: VHF
- [B]: UHF (329.15 to 335.00 MHz)
- [C]: SHF
- [D]: LF

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Glide Path operates in the UHF band, while the Localizer operates in the VHF band.

Q3: What is the audio modulation frequency and visual color of the ILS OUTER MARKER (OM)?

- [A]: 1300 Hz amber
- [B]: 400 HZ with BLUE light flashing dashes (two per second)
- [C]: 3000 Hz white
- [D]: Continuous tone red

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Outer Marker is identified by a 400 Hz tone keyed as low-pitched dashes with a flashing blue light.